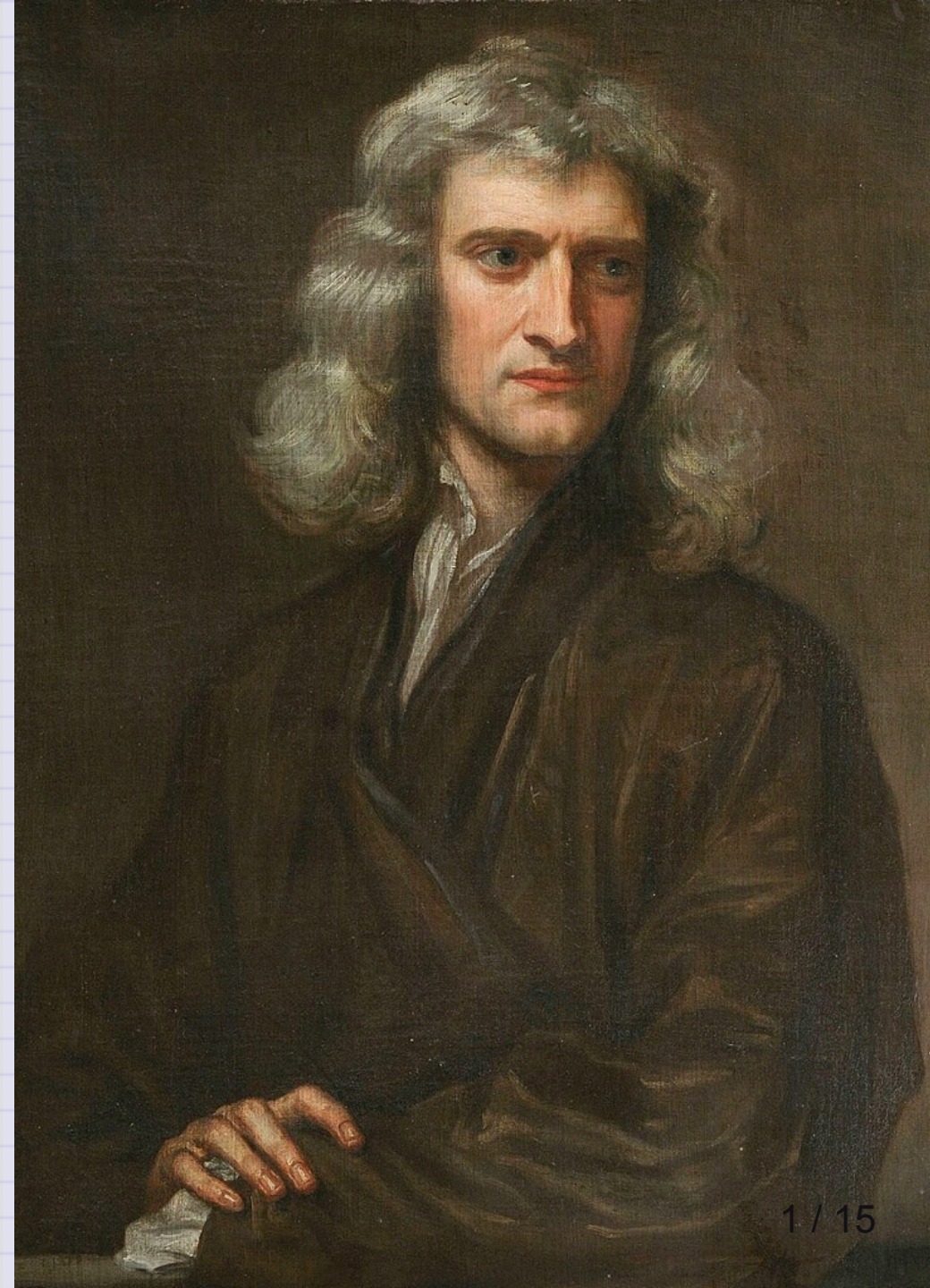


Day 02 - Newton's Laws

$$\mathbf{F}_{net} = m\mathbf{a}$$

Not shown in this picture of Newton are the countless illiterate mechanics and farmers, immigrant laborers, indigenous scholars, and other non elite members of society upon whose backs and accomplishments Newton's Principia was written.



Announcements

- Homework 1 is due next Friday
- Help sessions will start next week
- Friday's class will include AI policy discussion
 - We will get folks started with VS Code
 - We will also work Homework 1 Exercise 3 together

Goals for this week

Be able to answer the following questions.

- What is Classical Mechanics?
- How can we formulate it?
- What are the essential physics models for single particles?
- What mathematics do we need to get started?

Take 2 min to write down what comes to mind when asked:

What is "Classical" Physics?

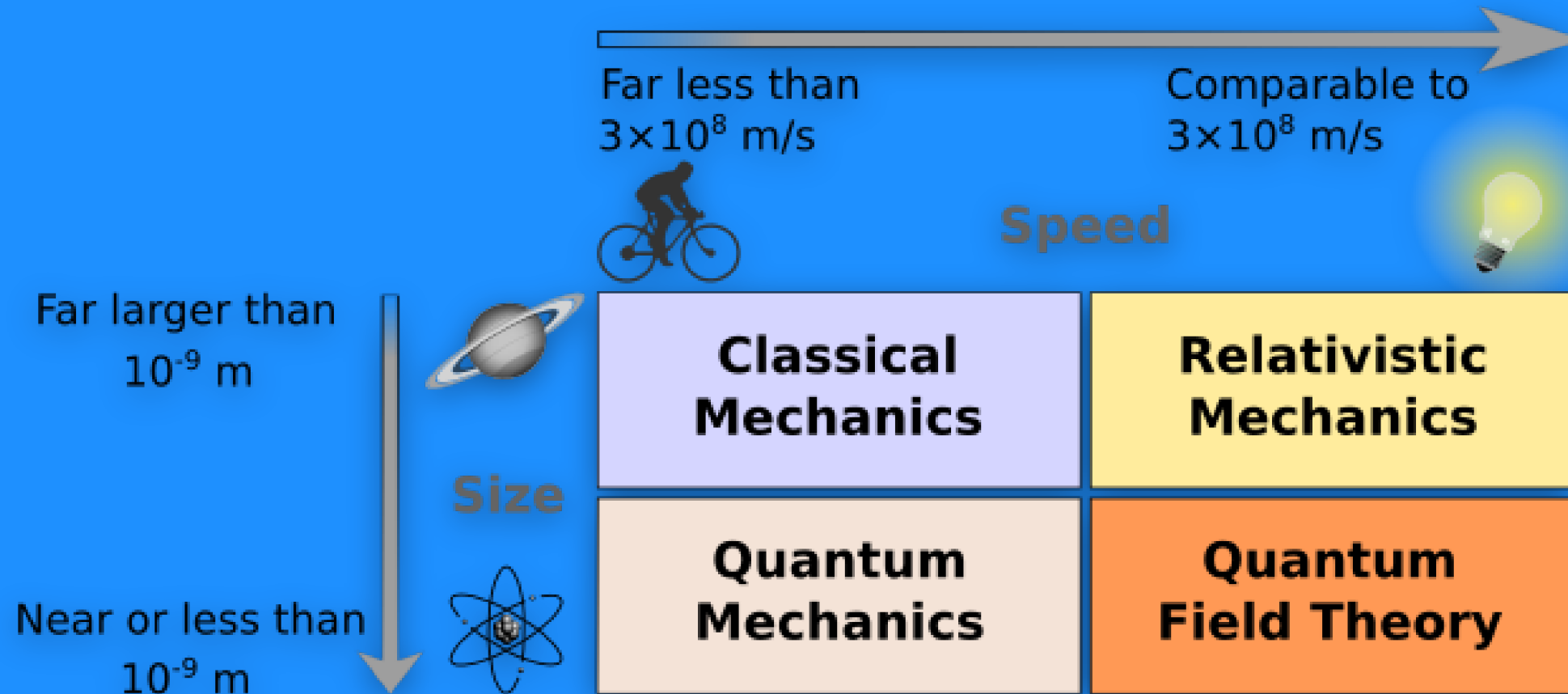
Classical Mechanics

Modeling large, slow-moving objects

Newton's Laws are but one of a number of formulations:

- Lagrangian Mechanics
- Hamiltonian Mechanics
- Dynamical Systems Theory
- ...

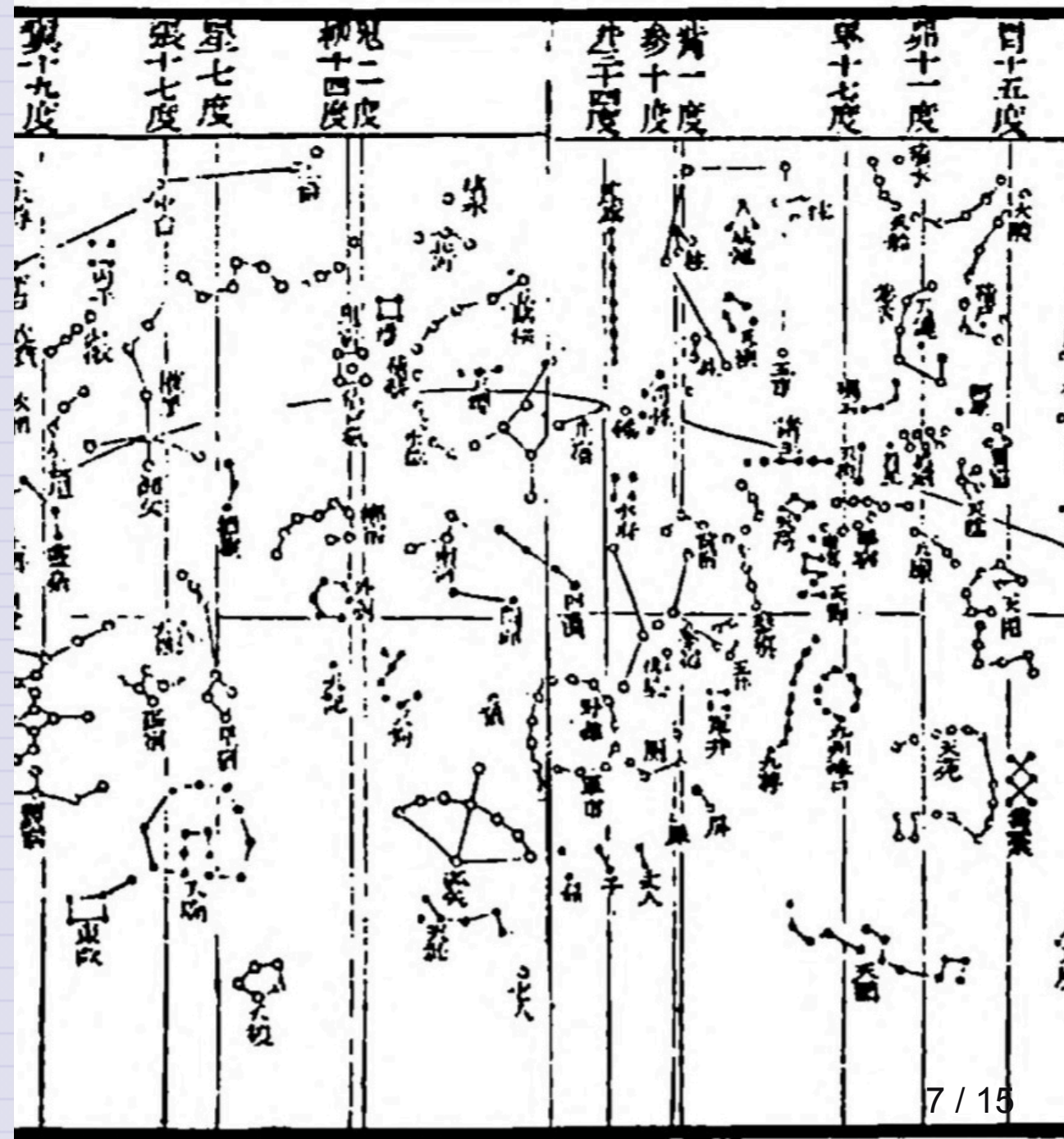
An Overview of Different Physics



Classical Mechanics has a long history

Wherever folks were figuring out their world, there was classical mechanics

- Astronomical analyses in Sub-Saharan Africa in 300 BCE
- Scientific expansion in China during the Song dynasty
- The Islamic Golden Age
- Indigenous astronomy



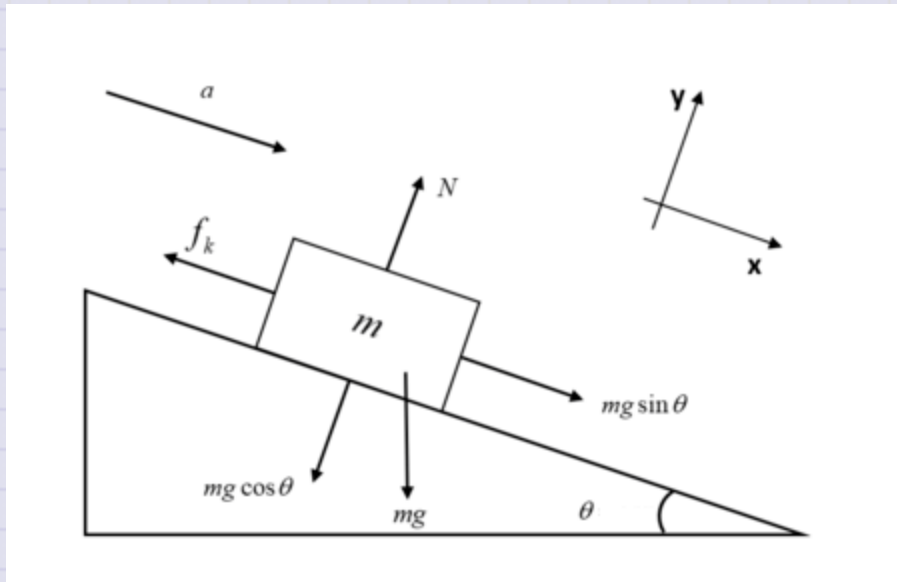
Classical Mechanics is still very relevant

Tiny Limbs and Long Bodies: Coordinating Lizard Locomotion
Research Lab



Source: <https://youtu.be/Qme07fA3Fj4>

Canonical Example from Introductory Physics



- Box on a ramp with a frictional interaction
- At what angle does it slide for a given μ_s ?

Think-Pair-Share

We used a tilted coordinate system ($x - y$ plane) to analyze the motion of a block on an inclined plane. How can we check that we did the gravitational force decomposition correctly?

Recall:

- $F_{\text{gravity}, x} = mg \sin(\theta)$
- $F_{\text{gravity}, y} = mg \cos(\theta)$

Come up with at least two checks.

Clicker Question 2-1

The formal definition of a Taylor series expansion around a point a is:

$$f(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \frac{f'''(a)}{3!}(x - a)^3 + \dots$$

This formula makes me feel:

1. Confident, I got this.
2. A little nervous, but I think I remember.
3. Uncomfortable, I don't remember this.
4. I have no idea what this is.

Think-Pair-Share

We derived the following differential equation for the falling ball in one-dimension:

$$\frac{d^2y}{dt^2} = +g - \frac{b}{m} \frac{dy}{dt} - \frac{c}{m} \left(\frac{dy}{dt} \right)^2$$

Let's assume the turbulent drag term is negligible. Is there an anti-derivative of the right-hand side of this equation? If so, what is it?

$$\frac{dv}{dt} = +g - \frac{b}{m} v$$

Example: Ball Falling in 1D in Air

We derived the following differential equation for the motion of a ball falling in air:

$$m\ddot{y} = mg - bv - cv^2$$

We argued for low speeds, we neglect the v^2 term.

$$m\ddot{y} = mg - bv$$

We can instead write this differential equation for v :

$$\dot{v} = g - \frac{b}{m}v$$

Example: Ball Falling in 1D in Air

Is this integrable? **Yes!**

$$\frac{dv}{dt} = g - \frac{b}{m}v$$

$$\frac{dv}{g - \frac{b}{m}v} = dt$$

$$\int \frac{dv}{g - \frac{b}{m}v} = \int dt$$

We will come back to this next week.

Vector Properties

Newtonian Mechanics is a vector theory. Here are a few mathematical properties of vectors:

- **Addition:** $\mathbf{A} + \mathbf{B} = (A_x + B_x)\hat{x} + (A_y + B_y)\hat{y} + (A_z + B_z)\hat{z}$
- **Scalar Multiplication:** $c\mathbf{A} = \langle cA_x, cA_y, cA_z \rangle$
- **Dot Product:** $\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z = AB \cos \theta$
- **Cross Product:** $\mathbf{A} \times \mathbf{B} = \langle A_y B_z - A_z B_y, A_z B_x - A_x B_z, A_x B_y - A_y B_x \rangle$
- **Unit Vectors:** $\hat{\mathbf{A}} = \frac{\mathbf{A}}{|\mathbf{A}|} \quad |\hat{\mathbf{A}}| = 1$